PCA Visualization Explorer

Machine Learning Assignment 12

October 7, 2026

1 Introduction

Principal Component Analysis (PCA) is a powerful dimensionality reduction technique. In this assignment, we use PCA to compress the high-dimensional Breast Cancer dataset (which contains 30 distinct features) down to just 2 principal components. The goal is to project the data into a 2D space so it can be visually analyzed by a human, while preserving as much of the original mathematical structure as possible.

2 2D PCA Visualization

Below is the scatter plot of the dataset projected onto its first two principal components. The points are colored based on their actual target labels (*malignant* vs *benign*).

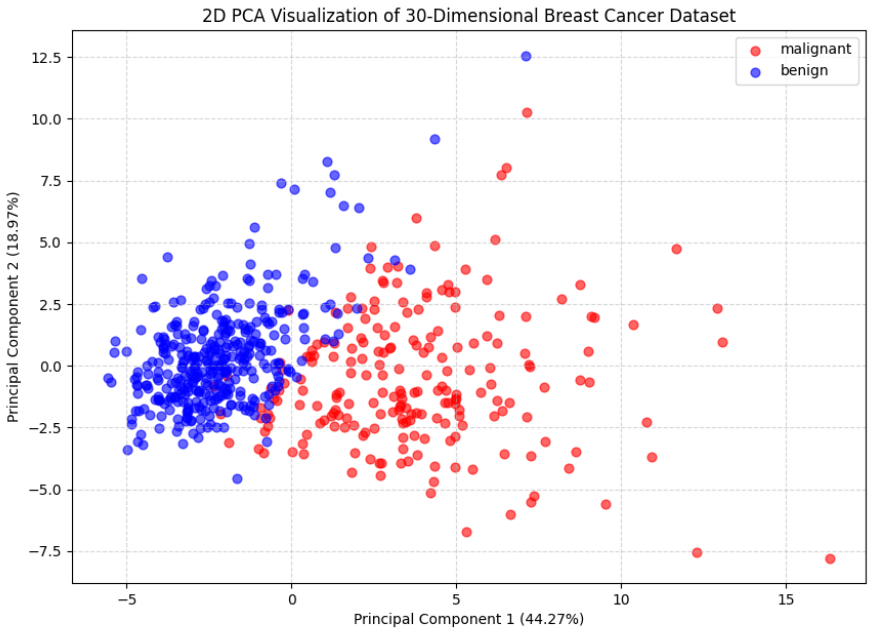


Figure 1: 2D PCA Projection of the 30-Dimensional Breast Cancer Dataset.

Despite discarding 28 dimensions, we can clearly see that the two classes naturally separate into two distinct clusters. This visual confirmation proves that the majority of the information needed to distinguish between malignant and benign tumors is contained within just these two mathematical vectors.

3 Variance Retained and Practical Interpretation

Quantitative Variance Reporting

When we reduce the data from 30 dimensions to 2, we inevitably lose some information. We quantify the remaining information as the "variance explained":

- **Principal Component 1 (PC1) explains:** 44.27% of the variance
- **Principal Component 2 (PC2) explains:** 18.97% of the variance
- **Total Variance Retained:** 63.24%

What does this mean in practice?

In practice, a total retained variance of $\sim$63% means that **nearly two-thirds of the total variation and structural information hidden across 30 different columns has been successfully compressed into just 2 columns**.

This highlights a fundamental trade-off in data science: **Simplicity vs. Information Loss**. By sacrificing 36.76% of the variance (information loss), we gained the ability to plot the entire dataset on a standard X-Y graph (simplicity). If we were to train a classifier (like Logistic Regression) solely on these 2 components, it would train exceptionally fast and be highly immune to the "curse of dimensionality", while still achieving a very high accuracy because the clusters are still well separated in this 63% variance subspace.